

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Lochana Sri R S /192521348 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lochana Sri R S

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWERS

1. CLOUD AUTO-SCALING AND LOAD BALANCING:

Introduction

A company experiences sudden traffic spikes during seasonal sales such as Black Friday or festive offers. Cloud computing helps manage this demand using auto-scaling, load balancing, and monitoring tools.

How Auto-Scaling and Load Balancing Handle Demand

- Auto-scaling automatically increases or decreases the number of servers based on traffic.
- During high traffic, additional virtual machines are created automatically.
- When traffic reduces, extra servers are removed to save costs.
- Load balancing distributes incoming user requests evenly across all available servers.
- This prevents any single server from becoming overloaded.

Reducing Downtime and Improving User Experience

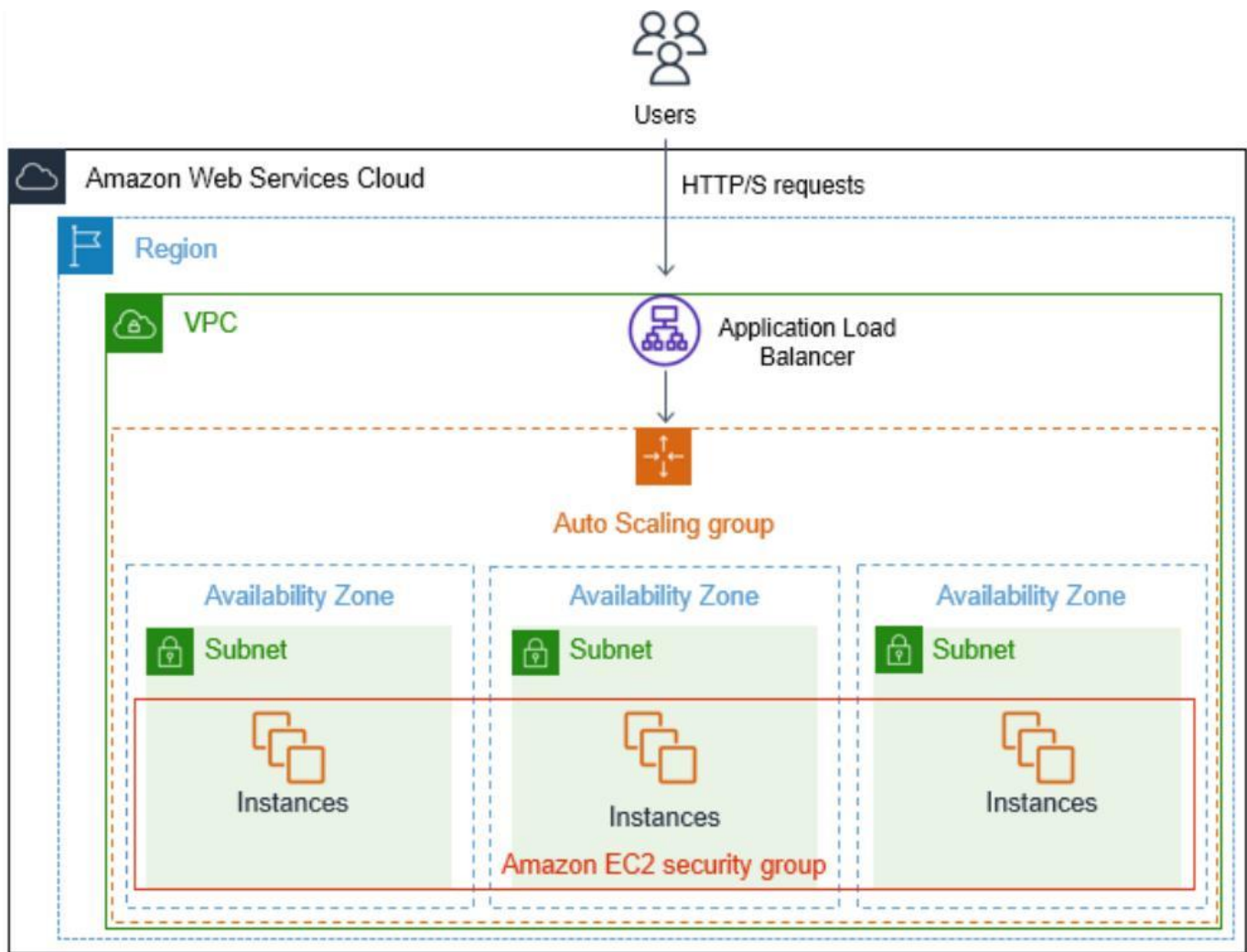
- Even if one server fails, the load balancer redirects traffic to healthy servers.
- Users experience faster page loading and uninterrupted service.
- Applications remain available even during heavy traffic.

Role of Monitoring Tools

Monitoring tools continuously track:

- CPU utilization
- Memory usage
- Network traffic
- Response time
- Server health

If resource usage exceeds a predefined threshold, monitoring tools trigger auto-scaling automatically.



Real-World Example

During the Amazon Great Indian Festival, millions of customers visit the website simultaneously. Auto-scaling launches additional cloud servers, while load balancers distribute customer requests evenly. Monitoring tools continuously observe system performance, ensuring the website remains responsive and available throughout the sale.

2. CLOUD STORAGE FOR SECURE BACKUP

Introduction

Organizations generate huge amounts of backup data that must be stored securely and remain available even if hardware failures occur. Cloud storage provides reliable, scalable, and cost-effective backup solutions.

Durability and Data Redundancy

- Cloud providers store multiple copies of data in different locations.
- If one storage device or data center fails, another copy is available.
- This ensures high durability and prevents data loss.

Encryption and Access Control

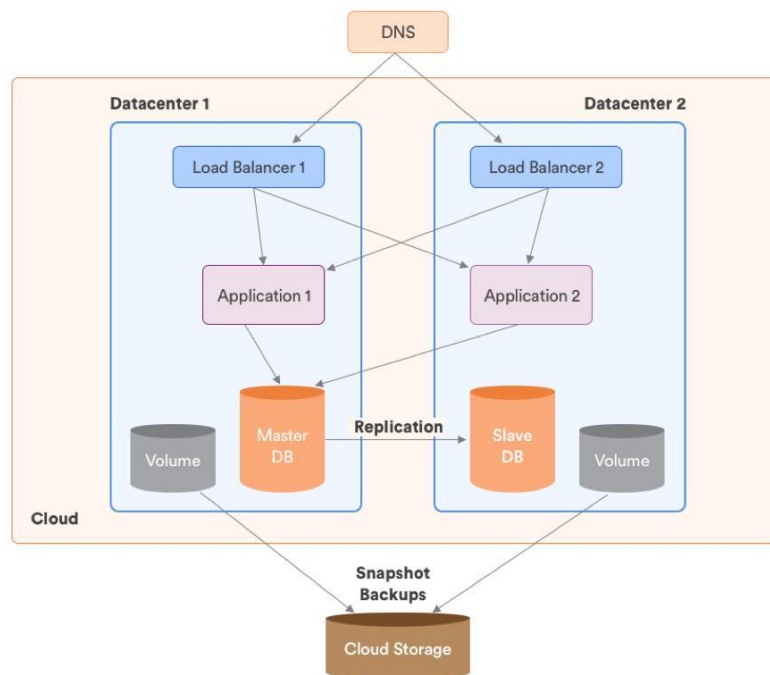
Cloud providers secure backup data using:

- Encryption at rest to protect stored data.
- Encryption in transit while transferring data.
- Role-Based Access Control (RBAC) to allow only authorized users.
- Multi-Factor Authentication (MFA) for additional security.

Cost Advantages

Compared to traditional storage systems:

- No need to purchase expensive storage hardware.
- No maintenance or infrastructure costs.
- Pay only for the storage used.
- Storage capacity can be increased or decreased easily.



Disaster Recovery Example

A hospital stores patient records in cloud backup. If a flood damages the local data center, all records can be restored quickly from cloud storage, allowing hospital operations to continue without losing critical medical information.

3. Global Deployment Using CDN

Introduction

A mobile application company wants users worldwide to access its services quickly. Cloud providers use Content Delivery Networks (CDNs) to improve performance.

How CDNs Improve Performance

- CDNs store copies of static content such as images, videos, and files in multiple locations worldwide.
- Users receive data from the nearest server instead of the main data center.
- This significantly reduces loading time.

Role of Edge Servers

Edge servers are located near users.

They:

- Reduce network latency.
- Deliver content faster.
- Reduce traffic on the main server.
- Improve application responsiveness.

Global Availability

Cloud providers maintain data centers across different countries.

Features include:

- Automatic traffic routing.
- Regional failover.

- High availability.
- Global load balancing.

Real-World Example

Netflix uses CDNs to stream movies globally. When a user watches a movie, the video is delivered from the nearest edge server instead of a distant central server, reducing buffering and providing smooth playback.

4. Cloud Security and Regulatory Compliance

Introduction

Financial institutions handle confidential customer information and must follow strict government regulations. Cloud providers offer advanced security and compliance features.

Regulatory Compliance and Auditing

Cloud providers support standards such as:

- ISO 27001
- PCI DSS
- GDPR
- SOC 2

They provide audit reports and compliance certifications to help organizations meet legal requirements.

Role of Logging, Monitoring, and Encryption

- Logging records every system activity and user action.
- Monitoring detects suspicious behavior and security threats.
- Encryption protects sensitive financial data from unauthorized access.

Together, these features improve overall cloud security.

Data Sovereignty

Some regulations require customer data to remain within a specific country.

Cloud providers allow organizations to:

- Select storage regions.
- Store data in approved locations.
- Ensure compliance with national data protection laws.



Example

A bank processes online fund transfers using encrypted cloud services. All transaction logs are securely recorded for auditing, while customer data remains stored in local data centers to comply with government regulations.

5. Containers and Kubernetes

Introduction

Modern software development uses containers to deploy applications efficiently across different environments.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Startup Time	Seconds	Minutes
Resource Usage	Lightweight	Heavy
Operating System	Shares host OS	Separate OS for each VM
Performance	Faster	Slightly slower
Scalability	Easy	More resource-intensive

Role of Kubernetes

Kubernetes automates:

- Container deployment.
- Scaling applications.
- Load balancing.
- Health monitoring.
- Automatic restart of failed containers.

This reduces manual effort and improves reliability.

Portability Across Clouds

Containers package the application and all its dependencies together.

Therefore, the same container can run on:

- Public cloud
- Private cloud
- Hybrid cloud
- Local systems

without modification.

Microservices Example

An online food delivery application can be divided into separate microservices such as:

- User Service
- Restaurant Service
- Order Service
- Payment Service
- Notification Service

Each service runs inside its own container. Kubernetes manages these containers independently, allowing the application to scale efficiently and remain highly available.

